In [ ]:
```
# EXPERIMENT 4 – DATA MANIPULATION WITH PANDAS
PIN NO.: 241UDS0110

## AIM
To perform data manipulation using Pandas: loading data, setting an index, selecting rows and columns, creating
calculated columns, filtering, sorting, updating values, renaming columns, and inspecting the final DataFrame.
```

In [ ]:
```
## Step 1 – Install and Import Pandas Library
```

In [1]:
```
import pandas as pd
```

In [ ]:
```
## Step 2 – Importing the Dataset
```

In [2]:
```
df_241UDS0110 = pd.read_csv("Experiment_4_241UDS0110_Fitness_Center_Sales.csv")
df_241UDS0110.head(5)
```

Out[2]: Columns 1–8 of 12

	PIN	Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price
0	241UDS0110	F001	Yoga Mat	Fitness Accessories	Vijayawada	34	850
1	241UDS0110	F002	Dumbbell Set	Strength Equipment	Hyderabad	21	2200
2	241UDS0110	F003	Skipping Rope	Cardio Equipment	Visakhapatnam	57	320
3	241UDS0110	F004	Resistance Bands	Fitness Accessories	Rajahmundry	46	450
4	241UDS0110	F005	Treadmill	Cardio Equipment	Pune	8	28500

Columns 9–12 of 12 · first column repeated

	Discount_Percent	Customer_Rating	Stock_Available	Payment_Method
0	5.0	4.4	22	UPI
1	8.0	4.6	12	Card
2	4.0	4.2	45	Cash
3	10.0	4.5	31	UPI
4	7.0	4.7	5	Card

In [ ]:
```
## Step 3 – Set Index for the DataFrame
```

In [3]:
```
df_241UDS0110 = df_241UDS0110.set_index("Product_ID")
df_241UDS0110.head()
```

Out[3]: Columns 1–8 of 11

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F001	241UDS0110	Yoga Mat	Fitness Accessories	Vijayawada	34	850	5.0
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyderabad	21	2200	8.0
F003	241UDS0110	Skipping Rope	Cardio Equipment	Visakhapatnam	57	320	4.0
F004	241UDS0110	Resistance Bands	Fitness Accessories	Rajahmundry	46	450	10.0
F005	241UDS0110	Treadmill	Cardio Equipment	Pune	8	28500	7.0

Columns 9–11 of 11 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method
F001	4.4	22	UPI
F002	4.6	12	Card
F003	4.2	45	Cash
F004	4.5	31	UPI
F005	4.7	5	Card

In [ ]:
```
## Step 4 – Info of the DataFrame
```

In [4]:
```
df_241UDS0110.info()
```

```
<class 'pandas.DataFrame'>
Index: 30 entries, F001 to F030
Data columns (total 10 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   PIN               30 non-null     str
 1   Product_Name      30 non-null     str
 2   Category          30 non-null     str
 3   City              30 non-null     str
 4   Units_Sold        30 non-null     int64
 5   Unit_Price        30 non-null     int64
 6   Discount_Percent  30 non-null     float64
 7   Customer_Rating   30 non-null     float64
 8   Stock_Available   30 non-null     int64
 9   Payment_Method    30 non-null     str
dtypes: float64(2), int64(3), str(5)
memory usage: 2.6+ KB
```

In [ ]:
```
## Step 5 – Columns & Rows Selection
### Selecting a Single Column
```

In [5]:
```
df_241UDS0110["Product_Name"]
```

Out[5]:
```
Product_ID
F001              Yoga Mat
F002          Dumbbell Set
F003         Skipping Rope
F004      Resistance Bands
F005             Treadmill
F006            Kettlebell
F007           Foam Roller
F008         Exercise Ball
F009      Adjustable Bench
F010        Cycling Helmet
F011       Stationary Bike
F012           Wrist Wraps
F013        Protein Shaker
F014    Weightlifting Belt
F015           Massage Gun
F016           Pull-up Bar
F017            Gym Gloves
F018        Rowing Machine
F019         Ankle Weights
F020          Sports Watch
F021           Shin Guards
F022         Tennis Racket
F023      Swimming Goggles
F024     Sports Duffel Bag
F025           Smart Scale
F026            Basketball
F027         Badminton Set
F028        Recovery Boots
F029     Sports Socks Pack
F030        Hydration Pack
Name: Product_Name, dtype: str
```

In [ ]:
```
## Step 6 – Selecting Multiple Columns
```

In [6]:
```
df_241UDS0110[["Product_Name", "Category", "City"]].head(3)
```

Out[6]:

Product_ID	Product_Name	Category	City
F001	Yoga Mat	Fitness Accessories	Vijayawada
F002	Dumbbell Set	Strength Equipment	Hyderabad
F003	Skipping Rope	Cardio Equipment	Visakhapatnam

In [ ]:
```
## Step 7 – Select Row Using iloc
```

In [7]:
```
df_241UDS0110.iloc[1]
```

Out[7]:
```
PIN                         241UDS0110
Product_Name               Dumbbell Set
Category            Strength Equipment
City                          Hyderabad
Units_Sold                           21
Unit_Price                         2200
Discount_Percent                    8.0
Customer_Rating                     4.6
Stock_Available                      12
Payment_Method                     Card
Name: F002, dtype: object
```

In [8]:
```
df_241UDS0110.loc["F030"]
```

Out[8]:
```
PIN                      241UDS0110
Product_Name         Hydration Pack
Category             Outdoor Sports
City                         Mumbai
Units_Sold                       20
Unit_Price                     1850
Discount_Percent                8.0
Customer_Rating                 4.4
Stock_Available                  15
Payment_Method                  UPI
Name: F030, dtype: object
```

In [9]:
```
df_241UDS0110.head(2)
```

Out[9]: Columns 1–8 of 11

	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
Product_ID							
F001	241UDS0110	Yoga Mat	Fitness Accessories	Vijayawada	34	850	5.0
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyderabad	21	2200	8.0

Columns 9–11 of 11 · first column repeated

	Customer_Rating	Stock_Available	Payment_Method
Product_ID			
F001	4.4	22	UPI
F002	4.6	12	Card

In [10]:
```
df_241UDS0110["Total_Sales_241UDS0110"] = df_241UDS0110["Units_Sold"] * df_241UDS0110["Unit_Price"]
df_241UDS0110.head(2)
```

Out[10]: Columns 1–8 of 12

	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
Product_ID							
F001	241UDS0110	Yoga Mat	Fitness Accessories	Vijayawada	34	850	5.0
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyderabad	21	2200	8.0

Columns 9–12 of 12 · first column repeated

	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110
Product_ID				
F001	4.4	22	UPI	28900
F002	4.6	12	Card	46200

In [ ]:
```
## Step 8 — Creating Discount Amount Column
```

In [11]:
```
df_241UDS0110["Discount_Amount_241UDS0110"] = df_241UDS0110["Total_Sales_241UDS0110"] *
df_241UDS0110["Discount_Percent"] / 100
df_241UDS0110.head(2)
```

Out[11]: Columns 1–8 of 13

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F001	241UDS0110	Yoga Mat	Fitness Accessories	Vijayawada	34	850	5.0
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyderabad	21	2200	8.0

Columns 9–13 of 13 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110
F001	4.4	22	UPI	28900	1445.0
F002	4.6	12	Card	46200	3696.0

In [ ]:
```
## Step 9 — Creating Net Sales Column
```

In [12]:
```
df_241UDS0110["Net_Sales_241UDS0110"] = df_241UDS0110["Total_Sales_241UDS0110"] -
df_241UDS0110["Discount_Amount_241UDS0110"]
df_241UDS0110.head(2)
```

Out[12]: Columns 1–8 of 14

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F001	241UDS0110	Yoga Mat	Fitness Accessories	Vijayawada	34	850	5.0
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyderabad	21	2200	8.0

Columns 9–14 of 14 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
F001	4.4	22	UPI	28900	1445.0	27455.0
F002	4.6	12	Card	46200	3696.0	42504.0

In [13]:
```
df_241UDS0110["Discount_Amount_241UDS0110"] = df_241UDS0110["Discount_Amount_241UDS0110"].round(2)
df_241UDS0110["Net_Sales_241UDS0110"] = df_241UDS0110["Net_Sales_241UDS0110"].round(2)
df_241UDS0110.head(2)
```

Out[13]: Columns 1–8 of 14

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F001	241UDS0110	Yoga Mat	Fitness Accessories	Vijayawada	34	850	5.0
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyderabad	21	2200	8.0

Columns 9–14 of 14 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
F001	4.4	22	UPI	28900	1445.0	27455.0
F002	4.6	12	Card	46200	3696.0	42504.0

In [ ]:
```
## Step 10 — Conditional Filtering
```

In [14]:
```
df_241UDS0110[df_241UDS0110["Units_Sold"] > 30].head(2)
```

Out[14]: Columns 1–8 of 14

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F001	241UDS0110	Yoga Mat	Fitness Accessories	Vijayawada	34	850	5.0
F003	241UDS0110	Skipping Rope	Cardio Equipment	Visakhapatnam	57	320	4.0

Columns 9–14 of 14 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
F001	4.4	22	UPI	28900	1445.0	27455.0
F003	4.2	45	Cash	18240	729.6	17510.4

In [15]:
```
df_241UDS0110[(df_241UDS0110["Total_Sales_241UDS0110"] > 30000) & (df_241UDS0110["Stock_Available"] < 25)].head(3)
```

Out[15]: Columns 1–8 of 14

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyderabad	21	2200	8.0
F005	241UDS0110	Treadmill	Cardio Equipment	Pune	8	28500	7.0
F009	241UDS0110	Adjustable Bench	Strength Equipment	Hyderabad	12	6800	11.0

Columns 9–14 of 14 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
F002	4.6	12	Card	46200	3696.0	42504.0
F005	4.7	5	Card	228000	15960.0	212040.0
F009	4.6	8	UPI	81600	8976.0	72624.0

In [16]:
```
df_241UDS0110[(df_241UDS0110["City"] == "Hyderabad") | (df_241UDS0110["City"] == "Mumbai")].head()
```

Out[16]: Columns 1–8 of 14

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyderabad	21	2200	8.0
F008	241UDS0110	Exercise Ball	Fitness Accessories	Mumbai	25	950	9.0
F009	241UDS0110	Adjustable Bench	Strength Equipment	Hyderabad	12	6800	11.0
F014	241UDS0110	Weightlifting Belt	Strength Equipment	Mumbai	16	1250	9.0
F017	241UDS0110	Gym Gloves	Fitness Accessories	Hyderabad	38	520	8.0

Columns 9–14 of 14 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
F002	4.6	12	Card	46200	3696.0	42504.0
F008	4.4	17	Card	23750	2137.5	21612.5
F009	4.6	8	UPI	81600	8976.0	72624.0
F014	4.6	11	Card	20000	1800.0	18200.0
F017	4.1	27	UPI	19760	1580.8	18179.2

In [17]:
```
df_241UDS0110[(df_241UDS0110["Customer_Rating"] ⩾ 4.5) & (df_241UDS0110["Stock_Available"] < 30)]
```

Out[17]: Columns 1–8 of 14

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyderabad	21	2200	8.0
F005	241UDS0110	Treadmill	Cardio Equipment	Pune	8	28500	7.0
F009	241UDS0110	Adjustable Bench	Strength Equipment	Hyderabad	12	6800	11.0
F010	241UDS0110	Cycling Helmet	Cycling Equipment	Delhi	31	1450	6.0
F011	241UDS0110	Stationary Bike	Cardio Equipment	Vijayawada	11	18500	8.0
F014	241UDS0110	Weightlifting Belt	Strength Equipment	Mumbai	16	1250	9.0
F015	241UDS0110	Massage Gun	Recovery Equipment	Bengaluru	14	4200	12.0
F018	241UDS0110	Rowing Machine	Cardio Equipment	Delhi	6	32000	10.0
F020	241UDS0110	Sports Watch	Wearable Devices	Chennai	17	5600	7.0
F022	241UDS0110	Tennis Racket	Racquet Sports	Mumbai	15	3100	8.0
F025	241UDS0110	Smart Scale	Wearable Devices	Hyderabad	13	2400	10.0
F027	241UDS0110	Badminton Set	Racquet Sports	Visakhapatnam	36	1150	7.0
F028	241UDS0110	Recovery Boots	Recovery Equipment	Bengaluru	5	14500	12.0

Columns 9–14 of 14 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
F002	4.6	12	Card	46200	3696.0	42504.0
F005	4.7	5	Card	228000	15960.0	212040.0
F009	4.6	8	UPI	81600	8976.0	72624.0
F010	4.5	23	Cash	44950	2697.0	42253.0
F011	4.8	6	Card	203500	16280.0	187220.0
F014	4.6	11	Card	20000	1800.0	18200.0
F015	4.7	9	UPI	58800	7056.0	51744.0
F018	4.8	4	Card	192000	19200.0	172800.0
F020	4.5	10	Card	95200	6664.0	88536.0
F022	4.6	9	Card	46500	3720.0	42780.0
F025	4.7	7	Card	31200	3120.0	28080.0
F027	4.5	28	UPI	41400	2898.0	38502.0
F028	4.8	3	Card	72500	8700.0	63800.0

In [18]:
```
df_241UDS0110[df_241UDS0110["City"].isin(["Hyderabad", "Bengaluru", "Pune"])].head()
```

Out[18]: Columns 1–8 of 14

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyderabad	21	2200	8.0
F005	241UDS0110	Treadmill	Cardio Equipment	Pune	8	28500	7.0
F007	241UDS0110	Foam Roller	Recovery Equipment	Bengaluru	29	780	5.0
F009	241UDS0110	Adjustable Bench	Strength Equipment	Hyderabad	12	6800	11.0
F015	241UDS0110	Massage Gun	Recovery Equipment	Bengaluru	14	4200	12.0

Columns 9–14 of 14 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
F002	4.6	12	Card	46200	3696.0	42504.0
F005	4.7	5	Card	228000	15960.0	212040.0
F007	4.1	19	Cash	22620	1131.0	21489.0
F009	4.6	8	UPI	81600	8976.0	72624.0
F015	4.7	9	UPI	58800	7056.0	51744.0

In [19]:
```
df_241UDS0110[df_241UDS0110["Category"].isin(["Fitness Accessories", "Strength Equipment"])].head()
```

Out[19]: Columns 1–8 of 14

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F001	241UDS0110	Yoga Mat	Fitness Accessories	Vijayawada	34	850	5.0
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyderabad	21	2200	8.0
F004	241UDS0110	Resistance Bands	Fitness Accessories	Rajahmundry	46	450	10.0
F006	241UDS0110	Kettlebell	Strength Equipment	Chennai	18	1650	6.0
F008	241UDS0110	Exercise Ball	Fitness Accessories	Mumbai	25	950	9.0

Columns 9–14 of 14 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
F001	4.4	22	UPI	28900	1445.0	27455.0
F002	4.6	12	Card	46200	3696.0	42504.0
F004	4.5	31	UPI	20700	2070.0	18630.0
F006	4.3	14	UPI	29700	1782.0	27918.0
F008	4.4	17	Card	23750	2137.5	21612.5

In [20]:
```
df_241UDS0110[df_241UDS0110["Payment_Method"].isin(["UPI", "Card"])].head()
```

Out[20]: Columns 1–8 of 14

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F001	241UDS0110	Yoga Mat	Fitness Accessories	Vijayawada	34	850	5.0
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyderabad	21	2200	8.0
F004	241UDS0110	Resistance Bands	Fitness Accessories	Rajahmundry	46	450	10.0
F005	241UDS0110	Treadmill	Cardio Equipment	Pune	8	28500	7.0
F006	241UDS0110	Kettlebell	Strength Equipment	Chennai	18	1650	6.0

Columns 9–14 of 14 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
F001	4.4	22	UPI	28900	1445.0	27455.0
F002	4.6	12	Card	46200	3696.0	42504.0
F004	4.5	31	UPI	20700	2070.0	18630.0
F005	4.7	5	Card	228000	15960.0	212040.0
F006	4.3	14	UPI	29700	1782.0	27918.0

In [21]:
```
df_241UDS0110.sort_values("Units_Sold", ascending=False).head()
```

Out[21]: Columns 1–8 of 14

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F003	241UDS0110	Skipping Rope	Cardio Equipment	Visakhapatnam	57	320	4.0
F013	241UDS0110	Protein Shaker	Fitness Accessories	Rajahmundry	52	280	7.0
F029	241UDS0110	Sports Socks Pack	Sports Apparel	Chennai	48	350	5.0
F004	241UDS0110	Resistance Bands	Fitness Accessories	Rajahmundry	46	450	10.0
F012	241UDS0110	Wrist Wraps	Fitness Accessories	Chennai	43	390	5.0

Columns 9–14 of 14 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
F003	4.2	45	Cash	18240	729.6	17510.4
F013	4.3	40	Cash	14560	1019.2	13540.8
F029	4.1	34	Cash	16800	840.0	15960.0
F004	4.5	31	UPI	20700	2070.0	18630.0
F012	4.0	36	UPI	16770	838.5	15931.5

In [22]:
```
df_241UDS0110.sort_values("Unit_Price", ascending=True).head()
```

Out[22]: Columns 1–8 of 14

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F013	241UDS0110	Protein Shaker	Fitness Accessories	Rajahmundry	52	280	7.0
F003	241UDS0110	Skipping Rope	Cardio Equipment	Visakhapatnam	57	320	4.0
F029	241UDS0110	Sports Socks Pack	Sports Apparel	Chennai	48	350	5.0
F012	241UDS0110	Wrist Wraps	Fitness Accessories	Chennai	43	390	5.0
F004	241UDS0110	Resistance Bands	Fitness Accessories	Rajahmundry	46	450	10.0

Columns 9–14 of 14 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
F013	4.3	40	Cash	14560	1019.2	13540.8
F003	4.2	45	Cash	18240	729.6	17510.4
F029	4.1	34	Cash	16800	840.0	15960.0
F012	4.0	36	UPI	16770	838.5	15931.5
F004	4.5	31	UPI	20700	2070.0	18630.0

In [23]:
```
df_241UDS0110.sort_values(["Category", "Units_Sold"], ascending=[True, False]).head()
```

Out[23]: Columns 1–8 of 14

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F024	241UDS0110	Sports Duffel Bag	Bags	Pune	24	1350	9.0
F026	241UDS0110	Basketball	Ball Sports	Delhi	29	760	4.0
F003	241UDS0110	Skipping Rope	Cardio Equipment	Visakhapatnam	57	320	4.0
F011	241UDS0110	Stationary Bike	Cardio Equipment	Vijayawada	11	18500	8.0
F005	241UDS0110	Treadmill	Cardio Equipment	Pune	8	28500	7.0

Columns 9–14 of 14 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
F024	4.4	16	UPI	32400	2916.0	29484.0
F026	4.3	21	Cash	22040	881.6	21158.4
F003	4.2	45	Cash	18240	729.6	17510.4
F011	4.8	6	Card	203500	16280.0	187220.0
F005	4.7	5	Card	228000	15960.0	212040.0

In [ ]:
```
## Step 11 – Updating Data in the Table
```

In [24]:
```
df_241UDS0110["City"] = df_241UDS0110["City"].replace(["Hyderabad"], ["Hyd"])
df_241UDS0110[df_241UDS0110["City"] == "Hyd"].head(2)
```

Out[24]:

Columns 1–8 of 14

Product_ID	PIN	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyd	21	2200	8.0
F009	241UDS0110	Adjustable Bench	Strength Equipment	Hyd	12	6800	11.0

Columns 9–14 of 14 · first column repeated

Product_ID	Customer_Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
F002	4.6	12	Card	46200	3696.0	42504.0
F009	4.6	8	UPI	81600	8976.0	72624.0

In [25]:
```
df_241UDS0110.loc["F001", "Unit_Price"] = 900
print(df_241UDS0110.loc["F001", "Unit_Price"])
```

```
900
```

In [26]:
```
df_241UDS0110 = df_241UDS0110.rename(columns={"Product_Name": "Item_Name", "City": "Location", "Customer_Rating": "Rating", "Net_Sales_241UDS0110": "Revenue"})
df_241UDS0110.head(1)
```

Out[26]:

Columns 1–8 of 14

Product_ID	PIN	Item_Name	Category	Location	Units_Sold	Unit_Price	Discount_Percent
F001	241UDS0110	Yoga Mat	Fitness Accessories	Vijayawada	34	900	5.0

Columns 9–14 of 14 · first column repeated

Product_ID	Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Revenue
F001	4.4	22	UPI	28900	1445.0	27455.0

In [27]:
```
df_241UDS0110.head()
```

Out[27]:

Columns 1–8 of 14

Product_ID	PIN	Item_Name	Category	Location	Units_Sold	Unit_Price	Discount_Percent
F001	241UDS0110	Yoga Mat	Fitness Accessories	Vijayawada	34	900	5.0
F002	241UDS0110	Dumbbell Set	Strength Equipment	Hyd	21	2200	8.0
F003	241UDS0110	Skipping Rope	Cardio Equipment	Visakhapatnam	57	320	4.0
F004	241UDS0110	Resistance Bands	Fitness Accessories	Rajahmundry	46	450	10.0
F005	241UDS0110	Treadmill	Cardio Equipment	Pune	8	28500	7.0

Columns 9–14 of 14 · first column repeated

Product_ID	Rating	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Revenue
F001	4.4	22	UPI	28900	1445.0	27455.0
F002	4.6	12	Card	46200	3696.0	42504.0
F003	4.2	45	Cash	18240	729.6	17510.4
F004	4.5	31	UPI	20700	2070.0	18630.0
F005	4.7	5	Card	228000	15960.0	212040.0

In [ ]:
```
## Step 28 — Final Dataset Information
```

In [28]:
```
df_241UDS0110.info()
```

```
<class 'pandas.DataFrame'>
Index: 30 entries, F001 to F030
Data columns (total 13 columns):
 #   Column                      Non-Null Count  Dtype  
---  ------                      --------------  -----  
 0   PIN                         30 non-null     str    
 1   Item_Name                   30 non-null     str    
 2   Category                    30 non-null     str    
 3   Location                    30 non-null     str    
 4   Units_Sold                  30 non-null     int64  
 5   Unit_Price                  30 non-null     int64  
 6   Discount_Percent            30 non-null     float64
 7   Rating                      30 non-null     float64
 8   Stock_Available             30 non-null     int64  
 9   Payment_Method              30 non-null     str    
 10  Total_Sales_241UDS0110      30 non-null     int64  
 11  Discount_Amount_241UDS0110  30 non-null     float64
 12  Revenue                     30 non-null     float64
dtypes: float64(4), int64(4), str(5)
memory usage: 4.3+ KB
```

In [ ]:
```
## RESULT
Pandas operations were performed on the fitness-equipment sales dataset, including row and column selection,
calculated sales columns, conditional filtering, sorting, updates, column renaming, and final DataFrame
inspection.
```